

Total No. of Questions : 6]

SEAT No. :

P4897

[Total No. of Pages : 2

**T.E./Insem. - 148**

**T.E. (Computer Engineering)**

**DATA COMMUNICATION AND WIRELESS SENSOR NETWORK**

**(2012 Pattern) (Semester - I)**

*Time : 1 Hour]*

*[Max. Marks : 30*

*Instructions to the candidates:*

- 1) Solve Que. 1 or 2, Que.3 or 4, Que. 5 or 6.*
- 2) Neat diagrams must be drawn wherever necessary.*
- 3) Assume suitable data if necessary.*
- 4) Figures to the right indicate full marks.*

**Q1) a)** Encode the following binary data stream into polar (RZ, NRZ), Manchester and differential Manchester codes for given Data stream : 11100101. **[5]**

**b)** Give definitions: **[3]**

i) Baud rate.

ii) Bit rate.

iii) SNR.

**c)** Write a note on Zigbee. **[2]**

OR

**Q2) a)** Explain what is meant by slope overload and granular noise distortion? Also explain how adaptive delta modulation improves system tolerance to slope overload. **[6]**

**b)** With the help of waveform, explain sampling, quantization and encoding. **[4]**

**P.T.O.**

- Q3)** a) Draw and explain diagram of Frequency Hopping Spread Spectrum (FHSS) and Direct Sequence Spread Spectrum (DSSS). [6]
- b) Consider Stop-and-wait ARQ system, the bandwidth of the line is 01 Mbps and 1 bit takes 20ms to make a round trip. What is the bandwidth-delay product? If the system data frames are 1000 bits in length. What is the utilization percentage of the link? [4]

OR

- Q4)** a) A pure ALOHA network transmits 200-bit frame on a shared channel of 200 kbps. What is the throughput if the system (all stations together) produces [4]
- i) 1000 frames per second.
- ii) 500 frames per second.
- b) Explain stop and wait ARQ, GO back-n ARQ and selective repeat ARQ. Comment on the performance of each. [6]

- Q5)** a) Explain block diagram of RFID reader and RFID Tag. [4]
- b) Define sensor network? Draw basic architecture and list its applications. [6]

OR

- Q6)** a) Explain typical sensing nodes architecture, how this sensing node is different from the nodes in other networks? [6]
- b) Explain the use of sensors in Robots. [2]
- c) What are advantages of RFID over bar code. [2]

